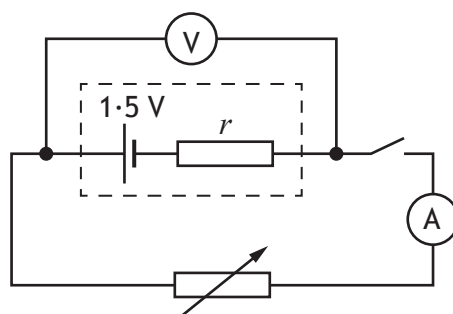


14. A student carries out an experiment, using the apparatus shown, to determine a value for the internal resistance r of a cell.



- (a) Describe how the student would use this apparatus, and analyse the data obtained, to determine the value for the internal resistance of the cell.

3



* X 8 5 7 7 6 0 1 3 6 *

14. (continued)

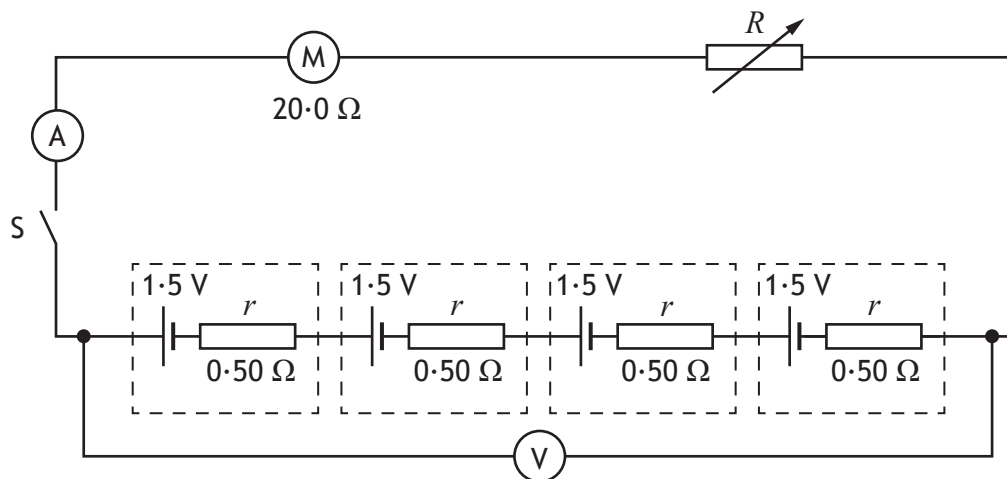
MARKS

DO NOT
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- (b) The internal resistance of the cell is determined to be 0.50Ω .

Four identical cells are now connected to a motor and a variable resistor as shown.

The EMF of each cell is 1.5 V .



- (i) State what is meant by an EMF of 1.5 V .

1

- (ii) Switch S is now closed. The reading on the ammeter is 0.20 A .

Determine the resistance R of the variable resistor.

4

Space for working and answer



* X 8 5 7 7 6 0 1 3 7 *

14. (continued)

- (c) The resistance of the variable resistor is now increased.
State what happens to the reading on the voltmeter.
Justify your answer.

3



* X 8 5 7 7 6 0 1 3 8 *